

UBT510: BIOLOGY FOR ENGINEERS

L	T	P	Cr
2	0	0	2.0

Course Objective: To learn about living world and basic functioning of biological systems. The course encompasses understanding of origin of life, its evolution and some of its central characteristics. It also aims to familiarize engineering students to some of the intricate biological phenomena and mechanisms.

Syllabus

Characteristics of life: Living versus non-living organisms, origin of life, theory of evolution, diversity of life, classification of life into animals, plants, fungi, protists, archaea and bacteria. Phylogenetics and its relationship with evolution.

Introduction to biological systems: Cell as basic unit of life, cellular organelles and their functions, important biomacromolecules (carbohydrates, lipids, proteins and nucleic acids) and their properties.

Cell membrane: Membrane structure, selective permeability, transport across cell membrane, active and passive transport, membrane proteins, type of transport proteins, channels and pumps, examples of membrane transport in cell physiology.

Classical and molecular genetics: Heredity and laws of genetics, genetic material and genetic information, Structure and properties of DNA, central dogma, replication of genetic information, universal codon system, encoding of genetic information via transcription and translation.

Laboratory Work: NA

After completion of this course the students will be able to:

1. Describe living-systems and differentiate them from non-living systems
2. Explain the theory of evolution and apply it non-living world
3. Apply properties of nucleic acids in molecular recognition-based diagnostics
4. Familiarized with various transport mechanisms across cell membranes

Explain how genetic information is stored, replicated and encoded in living

Recommended Books:

1. Nelson, D.L., Cox, M.M., Lehninger: Principles of Biochemistry, WH Freeman (2008) 5th ed.
2. Dhama, P.S., Srivastava, H.N. Chopra, G., A Textbook of Biology, Pradeep Publications (2008).
3. Das, H.K., Textbook of Biotechnology, John Wiley & Sons (2004) 3rd Edition.
4. Gardner, E.J., Simmons, M., Peter, S.D., Principles of Genetics, John Wiley & Sons (2008)
5. Albert, B., Essential Cell Biology, Taylor & Francis, London (2009)